1.4.3 Definition of Unit and its Types

The process of measurement is basically a process of comparison. To measure a quantity, we always compare it with some reference standard. For example, when we state that a rope is 10 meter long, it is to say that it is 10 times as long as an object whose length is defined as 1 metre. Such a standard is known as the unit of the quantity. Here 1 metre is the unit of the quantity 'length'.

An arbitrarily chosen standard of measurement of a quantity, which is accepted internationally is called unit of the quantity.

The units in which the fundamental quantities are measured are called *fundamental or base units* and the units of measurement of all other physical quantities, which can be obtained by a suitable multiplication or division of powers of fundamental units, are called *derived units*.

1.4.4 Different types of Measurement Systems

A complete set of units which is used to measure all kinds of fundamental and derived quantities is called a system of units. Here are the common system of units used in mechanics:

(a) **the f.p.s. system** is the British Engineering system of units, which uses ***foot***, ***pound*** and ***second*** as the three basic units for measuring length, mass and time respectively.

(b) **The c.g.s system** is the Gaussian system, which uses ***centimeter, gram*** and ***second*** as the three basic units for measuring length, mass and time respectively.

(c) **The m.k.s system** is based on ***metre, kilogram*** and ***second*** as the three basic units for measuring length, mass and time respectively.



The cgs, mks and SI are metric or decimal system of units. The fps system is not a metric system.

1.4.5 SI unit System

The system of units used by scientists and engineers around the world is commonly called ***the metric system*** but, since 1960, it has been known officially as the International System, or SI (the abbreviation for its French name, Système International). The SI with a standard scheme of symbols, units and abbreviations, were developed and recommended by the General Conference on Weights and Measures in 1971 for international usage in scientific, technical, industrial and commercial work. The advantages of the SI system are,

i) This system makes use of only one unit for one physical quantity, which means a rational system of units.

ii) In this system, all the derived units can be easily obtained from basic and supplementary units, which means it is a coherent system of units.

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